

**SYLLABUS :-**

**Objectives** To introduce the students to a systematic study of the mechanistic principles of fluid, and their applications to hydraulics as applicable to civil engineering.

**Topics** Class Hours Basic concepts applicable to fluid mechanics: Properties of Fluids- Mass density, specific weight, specific gravity, specific volume, vapour pressure, compressibility, elasticity, surface tension, capillarity; Newton's law of viscosity, classification of fluids, dynamic viscosity and kinematics viscosity, variation of viscosity with temperature

**Fluid Statics:** Measurement of Pressure, Pressure variation in a static fluid, PASCAL's law, Units and scales of pressure measurement -Atmospheric pressure, Absolute Pressure, Gauge Pressure and Vacuum Pressure, Hydrostatic Paradox. Piezometer, U-Tube Manometer, Single Column Manometer, U-Tube Differential Manometer, Inverted U Tube Differential Manometer, Micro-manometers. Mechanical Pressure Gauges. Hydrostatic force on plane and curved surface- Total Pressure and Center of Pressure, Pressure Diagram, Total Pressure on Plane Surfaces and Depth of Center of Pressure, Total Pressure on Curved Surfaces, Practical applications of Total Pressure and Center of Pressure

**Buoyancy and Flotation-** Buoyant force, Buoyancy and Center of Buoyancy, Archimedes Principle, Principle of Floatation. Metacentre and Metacentric Height, Equilibrium of Floating bodies and Submerged bodies. Evaluation of Metacentric Height -Theoretical Method and Experimental Method

**Fluid Kinematics:** Fluid Flow Methods of analysis of fluid motion, Streamlines, Path lines, Streak lines and Stream tubes. Types of fluid flow Steady and unsteady flow, Uniform and non-uniform flow, Laminar, Transitional and Turbulent flow Reynolds number, Reynolds Experiment, Rotational and Irrotational flow, Subcritical, Critical and Supercritical flow, Compressible and Incompressible Flow, One, Two and Three Dimensional Circulation and vorticity, Velocity potential and Stream function, Flow net

**Fluid Dynamics-** Euler's equation, Bernoulli's equation, Energy correction factor, momentum correction factor.

**Flow Measuring Devices-** Measurement of discharge Venturi meter, Orifice meter, Nozzle meter, Bend meter, Rotometer. Measurement of velocity -Pitot tube. Orifice - Classification, Flow through a Reservoir Opening, i.e., Orifice, Trajectory of free -jet, Hydraulic Coefficients, Experimental determination of hydraulic coefficient, Small and large orifice, Time of emptying a tank with orifice.

**Flow Past immersed bodies-** Drag and lift, Types of drag, Drag on a sphere, cylinder, flat plate

**Laminar Flow and Turbulent Flow:** Resistance to flow of fluid in smooth and rough pipes, Boundary layer theory. Boundary-layer thickness, displacement, momentum